TicketLab Architecture & Refactoring Plan

Direct Successor to conference-platform · Production Hardening Specification

Target: Production | Laravel 13 · Livewire 4 | Tooling: Mago Only

Executive Architectural Decisions (Locked)

TicketLab evolves from a single-event prototype into a **Universal Multi-Tenant Event & Ticketing Platform**. It inherits all proven features of `conference-platform` (speaker bios, ticket packages, dynamic questionnaires, batch email broadcasting, queue monitor) while adding multi-tenant boundary isolation, concurrency row locks, and modern reactive Livewire 4 UI.

- **Storage**: Local `public` disk with symlink; fully cloud-ready for S3/Cloudinary.
- **Dedicated Mode**: Interactive long-form home page with smooth-scroll sections + discrete direct marketing sub-pages.
- **QR Engine**: Scannable QR code generation linking to the live registration URL.
- **Queue & Emails**: Queued ticket invoice PDF generation + `Bus::batch()` reminder broadcast with `QueueMonitor` dashboard.
- **Surface Isolation**: Strict separation of 4 roles so staff, admins, and attendees never see internal controls they cannot use.

1. FOUR ISOLATED APPLICATION SURFACES

1. Public & Dedicated Surface

- Dedicated Conference Home / Marketplace
- Speaker Bio Drawers & Schedule Tabs
- Ticket Selection & Custom Questionnaire
- Paystack Checkout & Digital Pass View

2. Ground Ops Terminal

- Fullscreen Camera QR Scanner (Audio & Haptics)
- Walk-In Cash Desk Register (GHS Chips)
- Gate Turnout Counter Scoped to Active Event
- *Zero administrative or ledger links*

3. Tenant Admin Command

- Event Lifecycle CRUD & Active Event Switcher
- Staff & Usher Management CRUD (RBAC)
- Ticket Tiers, Perks & Roster CSV Stream
- Email Broadcast Hub & Queue Monitor
- Daily Session Financial Ledger & PDF Export

4. Platform Super Admin

- Global Platform GMV & Developer Fee Accumulator
- Tenant Organizations Lifecycle CRUD
- Runtime Personality Switcher (Platform vs Dedicated)
- Cross-Tenant Audit & Global System Heartbeat

SURFACE-01

Milestone 1: Public Navbar Purge & Surface Demarcation

Defect in Code: `floating-nav.blade.php:85-107` injects "Ops Hub" and "Scanner" into the public navbar when authenticated.

? Remove "Ops Hub" and "Scanner" from `components/floating-nav.blade.php`.

? For unauthenticated visitors: show Home, Schedule, Speakers, Packages, FAQs, and "Sign In".

? For staff/admin users: show a discrete "Manage Event ?" pill linking to `/admin` or `/ops`.

? Create dedicated `resources/views/layouts/admin.blade.php` with full administrative sidebar.

? Slim down `resources/views/layouts/ops.blade.php` to contain only Door Scanner and Cash Desk.

STAFF-01

Milestone 2: Staff & User Management CRUD

Defect in Code: Currently 0 user management interfaces exist in the application. Admins cannot add or manage ushers.

- ? Create `App\Livewire\Admin\UserManagement` at `/admin/users`.
- ? Table listing staff scoped to `auth()->user()->organization_id`.
- ? Sheaf UI Modal for Create & Edit: Name, Email, Password, Role (Admin, Staff, Usher), and Zone (Subdivision).
- ? Persist using `$user->assignMembership($role, $orgId, $subdivisionId)` to honor mass-assignment protection.
- ? Prevent admin self-deletion safety guard (`auth()->id() === $user->id`).

EVENT-01

Milestone 3: Event Lifecycle Management CRUD

Defect in Code: `git grep "Event::create" app/` yields 0 results. If no event is seeded, the Customizer Studio hits a dead-end.

- ? Create `App\Livewire\Admin\EventManager` at `/admin/events`.
- ? Modal/Form: Title, Slug (auto-generated), Venue, Start Date, End Date, Banner image upload, Status (draft/published), Visibility.
- ? Provide an Active Event Switcher in the Admin header, storing `session('admin_active_event_id')`.
- ? Automatically scope Customizer, TicketManager, Roster, Ledger, and Marketing Studio to the active event.

DEDICATED-01 Milestone 4: Dedicated Mode Parity with conference-platform

Target: Elevate `tenants/default-bespoke.blade.php` to match the full conference experience from `conference-platform`.

- ? **Hero with Live Countdown**: Date badge with Alpine.js countdown clock (Days, Hours, Mins, Secs) and smart CTA.
- ? **Animated Marquee**: Smooth CSS ticker ("Connect · Grow · Lead · Purposeful Careers · Mentorship · Leadership").
- ? **Conference Packages & Perks**: Dynamic cards showing price, availability, and perk bullet points.
- ? **Speaker Directory**: Speaker cards with photo, company badge, and click-to-open Emil Kowalski spring bio drawers.
- ? **Dedicated Sub-Pages**: `/about-us`, `/speakers`, `/conference-packages`, `/highlights`, `/partnership`, `/faq`, `/contact`.
- ? **Contact Form**: Working inquiry form dispatching `App\Jobs\SendEnquiryEmail`.

QUEUE-01 Milestone 5: Queued Ticket Emailing, Broadcast Hub & Queue Monitor

Target: Replicate `conference-platform`'s batchable email dispatch and queue monitoring infrastructure.

- ? **Queued Ticket PDF Email**: Create `App\Jobs\SendTicketEmail`. Render branded PDF ticket with QR pass via DomPDF, attach, and email to buyer on fulfillment.
- ? **Batch Reminder Broadcast**: Create `App\Jobs\SendEventReminders` (Batchable) and `App\Livewire\Admin\BroadcastHub` at `/admin/broadcasts`.
- ? Dispatch broadcasts using `Bus::batch($jobs)->name("Broadcast: {$subject}")->dispatch()`.
- ? **Queue Monitor Dashboard**: Port `App\Livewire\Admin\QueueMonitor` to `/admin/queue-monitor`. Real-time counts of pending, failed, and batch jobs with retry/flush actions.

STUDIO-01 Milestone 6: Real QR Engine & Marketing Studio Polish

Defect in Code: `marketing-studio.blade.php:420-494` draws fake pseudo-random Canvas squares and 3 static boxes.

- ? Install verified QR generator: `composer require chillerlan/php-qrcode`.
- ? Create backend QR data URI generator endpoint: `/api/qr?content={url}` producing scannable PNG/SVG data.
- ? Wire real QR code image into Canvas context; enable instant **Download Real SVG** and **Download Real PNG**.
- ? Implement Canvas multi-line text wrapping to eliminate title/venue text clipping across 1:1, 9:16, and 16:9 formats.
- ? Add custom background image upload for promotional flyers.

STUDIO-02 Milestone 7: Visual Customizer Studio Live Preview & Uploads

Defect in Code: Speaker photos are manual text strings with no file upload; no real-time live preview of color/hero edits.

- ? Add `Livewire\WithFileUploads` to `ManagesEventSpeakers`; store photos in `storage/app/public/speakers`.
- ? Add responsive side-by-side or drawer live preview updating instantly as palette or copy changes.

2. COMPONENT & ROUTE ARCHITECTURE MAP

Area	Component / Controller	Route Path	Layout Shell
Public	`App\Http\Controllers\HomeController`	`GET /`	`layouts/public.blade.php`
Public	`App\Livewire\Public\EventShow`	`GET /event/{org}/{slug}`	`layouts/public.blade.php`

Area	Component / Controller	Route Path	Layout Shell
Dedicated	Blade / Folio Sub-Pages	`GET /speakers, /conference-packages...`	`layouts/public.blade.php`
Ground Ops	`App\Livewire\Ops\ScannerTerminal`	`GET /ops/scanner`	`layouts/ops.blade.php`
Ground Ops	`App\Livewire\Ops\CashDeskTerminal`	`GET /ops/cash-desk`	`layouts/ops.blade.php`
Admin	`App\Livewire\Admin\EventManager`	`GET /admin/events`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\UserManagement`	`GET /admin/users`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\TicketManager`	`GET /admin/tickets`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\AttendeeRoster`	`GET /admin/attendees`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\BroadcastHub`	`GET /admin/broadcasts`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\QueueMonitor`	`GET /admin/queue-monitor`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\DailySessionLedger`	`GET /admin/sessions`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\EventCustomizer`	`GET /admin/studio`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\MarketingStudio`	`GET /admin/studio/marketing`	`layouts/admin.blade.php`
Admin	`App\Livewire\Admin\DataExportCentral`	`GET /admin/exports`	`layouts/admin.blade.php`
Super Admin	`App\Livewire\Platform\PlatformDashboard`	`GET /platform`	`layouts/platform.blade.php`